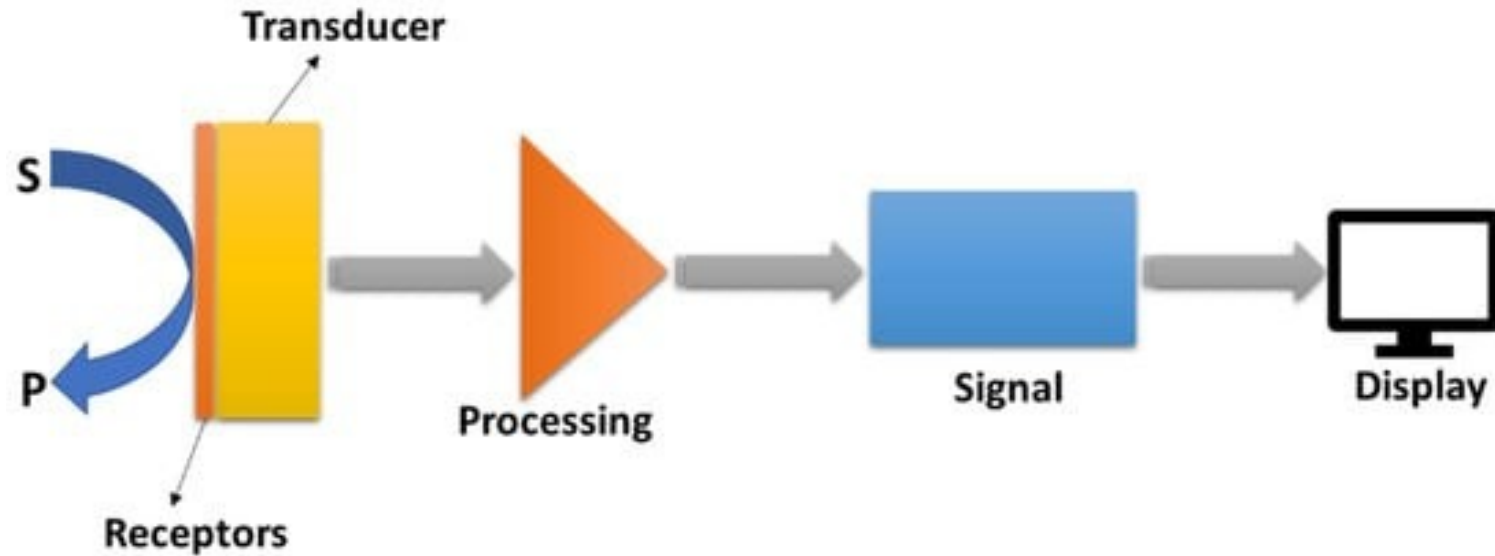


Electro-analytical sensors and their application in pharmaceutical analysis

- **Chemical sensor** is defined as “a device that converts chemical data into an **analytically usable signal**”
- a chemical sensor is constituted of two essential functional units:
 - i. A receptor and
 - ii. A physicochemical transducer.
- The receptors are variable and can range from **activated or doped surfaces** to **complex (macro)molecules** that create highly **specific interactions with the analyte.**



A schematic diagram illustrating the major components of a standard sensor.

- If the receptor is of biological origin (e.g., DNA, antibodies, and enzymes), the device is referred to as a **biosensor**.

- The receptor interacts with the analyte, converting the recognition event into a **predetermined output signal**.
- Another critical component of sensors is the **transducer**, which is responsible for converting the **signal created by the receptor–analyte** interaction into a **readable value**.
- In the case of electrochemical sensors, the **electrode acts as the transducer**.

Electrochemical Sensors

Electrochemical sensors are the most versatile and highly developed chemical sensors.

They are divided into several types:

- Potentiometric (measure voltage)
- Amperometric (measure current)
- Conductometric (measure conductivity)

In all these sensors, special electrodes are used.

Potentiometric sensors

- In potentiometric sensors, the analytical information is obtained by converting the recognition process into a **potential signal**,
- Which is logarithmically proportional to the concentration (activity) of species generated or consumed in the recognition event
- PH electrodes have been the most extensively used **potentiometric device**

Amperometric Sensors

- Amperometric measurements are frequently used as an analytical method of high accuracy and sensitivity.
- Applied voltage serves as a driving force for electrocatalytic **redox reactions** that generate **electrical currents**
- **Currents proportional to the concentration of the analyte**

Conductometric sensors

- **Conductometric sensors** are chemical sensors which measure the electrical conductivity of an analyte on the basis of the concentration of that analyte.

Biosensors

Biosensors are composed of two main parts:

- A Biological recognizing component (**enzyme, antibody, cell, tissue, etc.**)
- Directly connected to a **physical transducer** (voltammetric, amperometric, conductometric, spectrophotometric, etc)
- The physicochemical interaction between analyte and the bio-recognizing agent is monitored by a transducer.

- In the case of **electrochemical biosensors**, the signal detection occurs at the **electrode/solution interface**, which can be **dynamic or static**.
- In dynamic methods, i.e. voltammetry (amperometric biosensor), the interaction must involve a redox process followed by electron transfer.

- In static methods (potentiometry) however, the concentration of charged species, as a function of the **electrochemical potential**, is monitored by a **potentiometric biosensor**.
- The use of either biosensor depends on the characteristics of the analyte.
 - ✓ Thus, for amperometric biosensors, inorganic or organic species must undergo a redox process at working potentials, i.e. electroactive species,
 - ✓ While static methods are applicable to charged species.

- Among the most widely used recognizing agents are enzymes (e.g. peroxidases, phenoloxidases and sterases), antibodies, drug receptors, and DNA.
- The use of entire cells or tissues, as well as other components, is less common in biosensors applied to pharmaceutical analysis.

Biosensors in pharmaceutical analysis

- The majority of biosensors are based on:
 - **enzymatic reactions,**
 - **The nature of the products in the biochemical reaction,**
 - **Molecular characteristics of the analyte,**
- Concerning oxidizable drugs, the best choice would be amperometric transducers and the enzymes:
 - peroxidases, laccases, tyrosinase, superoxide dismutase

Anti-inflammatory and analgesic drugs

- A number of biosensors based on **Salicylate Hydroxylase (SH)** alone or coupled to other enzymes, i.e., glucose oxidase or tyrosinase.
- The analytical signal is obtained by **electron transfer** during the **oxireduction of cathecolic product**, which results from the **hydroxylation and decarboxylation of salicylate**.
- In the presence of **hydrogen peroxide**, the **Horseradish peroxidase (HRP)** catalyzes the oxidation of paracetamol (acetaminophen) into **N-acetyl-p benzoquinoneimine**.

Cont..

- The amperometric biosensors based transducer surface are the main analytical strategies used for **paracetamol analysis**.

Beta-lactam antibiotics

- The first biosensors for penicilline were based on penicillinase enzyme, a specific type of beta-lactamase.
- Beta-lactam ring hydrolysis generates **penicilloic acid**, which is detected by a **potentiometric sensor**.
- An amperometric biosensor developed by **tyrosinase electrodes** was evaluated for the indirect detection of **penicillin**.
- The enzymes oxidize catechol to quinone, which is electrochemically reduced.

- Penicillin was added to the same electrolytic cell which reacts with the **o-quinone producing the quinone,**
- **Consequently decreasing the analytical signal.**
 - **The decrease in current is proportional to the penicillin concentration**

Antineoplastic

- The interaction between DNA or RNA and antineoplastic drug can be electrochemically monitored
 - the suppression of anodic peaks corresponding to the oxidation of **purines and pyrimidines of nucleic acids.**
 - The magnitude of the alterations in the analytical signal can be associated to the drug concentration.
- (e.g. anodic peak suppression of adenine)

Imidazole compounds

- DNA sensors have been successfully used to study this class of compound.
- The electrochemical behavior of imidazolic compounds have been **studied by DNA modified electrodes**, not only for analytical purposes, but also in order to **elucidate the mechanism**, as well as **the level of toxicity**.

Tetracyclines and quinolones

- The principle of detection was based on the **potentiometric determination** of **carbon dioxide** produced by microbial activity of *Escherichia coli*.
- A quantitative assay for **ciprofloxacin**, a quinolinocarboxylic acid derivative based on HRP-biosensors.
- The principle of detection was based on electrochemical **reduction of quinones enzymatically**.

- Formed on the electrode surface, which was blocked by piperazine groups of ciprofloxacin.
- Hence, in this indirect method, the current suppression was proportional to the **ciprofloxacin amount**.

Neuroleptics and Antidepressants

- The phenothiazine derivatives were analyzed based on the use of **Horseradish peroxidase** and **carbon electrodes** coupled to FIA systems.
- The biosensor for MAO inhibitor analysis is based on monoamine oxidases (MAO) connected to **amperometric transducers**.
- Neuroleptic can quantified by **amperometric or potentiometric** techniques.

Conclusion

- Detection of substances can be achieved by the use of sensors.
- Biosensor is a promising tool that have low cost, simplicity of analysis, selectivity and good sensitivity.
- Biosensors use in the quality analysis of drugs, medicines, and other analytes of interest in the pharmaceutical area.
- Biosensors allow quantification not only of the active component in pharmaceutical formulations, but also the analysis of degradation products and metabolites in biological fluids.

THE END